Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

When forces are applied on a body such that it is still in static equilibrium, then the extent to which the body gets deformed, depends on

(A) nature of the material

(B) magnitude of deforming force

(C) Both (a) \& (b)

(D) None of these

Q.2

Which of the following affects the elasticity of a substance?

(A) Change in temperature

(B) Hammering and annealing

(C) Impurity in substance

(D) All of the above

Q.3

which of the following is correct for young's modulus of elasticity (γ)? [where $\mathrm{r}=$ radius of cross section of wire, $l=$ length of wire]

(A) $\gamma \propto \mathrm{r}^2$

(B) $\gamma \propto l^3$

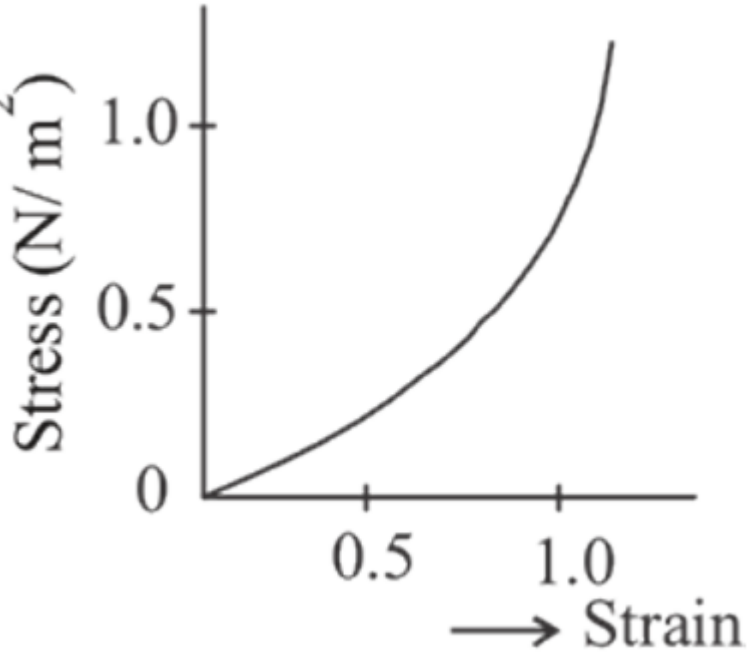
(C) $\gamma \propto 1/\mathrm{r}^2$

(D) $\gamma \propto l^2$

Q.4

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The graph given is a stress-strain curve for



(A) elastic objects

(B) plastics

(C) elastomers

(D) None of these Str

Q.5

For the same cross-sectional area and for a given load, the ratio of depressions for the beam of a square cross-section and circular cross-section is

(A) $3 : \pi$

(B) $\pi : 3$

(C) $1 : \pi$

(D) $\pi : 1$

Q.6

The length of an elastic string is a metre when the longitudinal tension is 4 N and b metre when the longitudinal tension is 5 N. The length of the string in metre when the longitudinal tension is 9 N is

(A) $a - b$

(B) $5b - 4a$

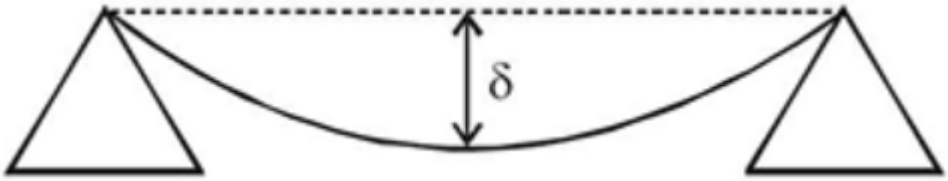
(C) $2b - \frac{1}{4}a$

(D) $4a - 3b$

Q.7

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A beam of metal supported at the two edges is loaded at the centre. The depression at the centre is proportional to



(A) Y^2

(B) Y

(C) $1/Y$

(D) $1/Y^2$

Q.8

For the same cross-sectional area and for a given load, the ratio of depressions for the beam of a square cross-section and circular cross-section is

(A) $3 : \pi$

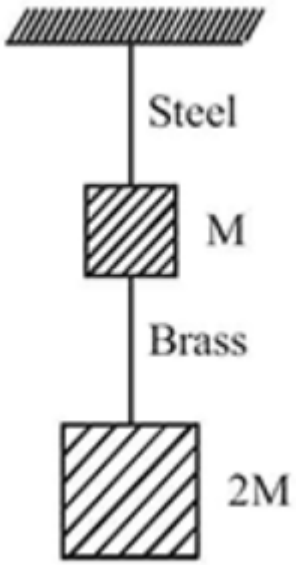
(B) $\pi : 3$

(C) $1 : \pi$

(D) $\pi : 1$

Q.9

If the ratio of lengths, radii and Young's moduli of steel and brass wires in the figure are a, b and c respectively, then the corresponding ratio of increase in their lengths is :



(A) $\frac{3c}{2ab^2}$

(B) $\frac{2a^2c}{b}$

(C) $\frac{3a}{2b^2c}$

(D) $\frac{2ac}{b^2}$

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Q.10

A uniformly tapering conical wire is made from material of Young's modulus Y and has a normal, unextended length L. The radii, at the upper and lower ends of this conical wire, have values R and $3R$, respectively. The upper end of the wire is fixed to a rigid support and a mass M is suspended fromits lower end. The equilibrium extended length, of this wire, would equal :

(A) $L\left(1+\frac{2}{9}\frac{Mg}{\pi YR^2}\right)$

(B) $L\left(1+\frac{1}{9}\frac{Mg}{\pi YR^2}\right)$

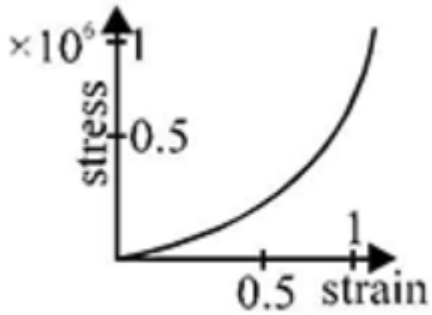
(C) $L\left(1+\frac{1}{3}\frac{Mg}{\pi YR^2}\right)$

(D) $L\left(1+\frac{2}{3}\frac{Mg}{\pi YR^2}\right)$

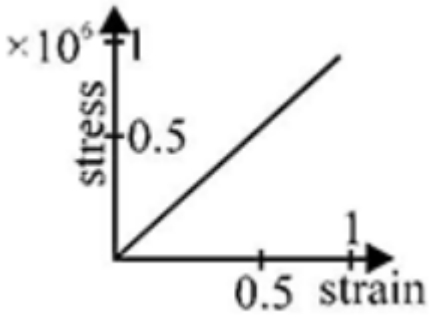
Q.11

Stress Vs strain for the elastic tissue of the aorta, the large tube (vessel) carrying blood from the heart, will be : [stress is proportional to square of the strain for the elastic tissue of the aorta]

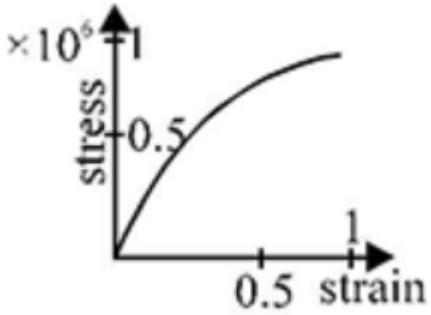
(A)



(B)

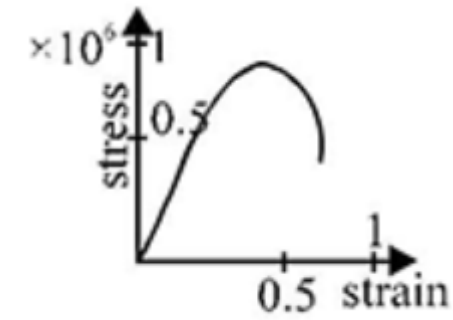


(C)



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(D)



Q.12

A metal wire of length L_1 and area of cross-section A is attached to a rigid support. Another metal wire of length L_2 and of the same cross-sectional area is attached to the free end of the first wire. A body of mass M is then suspended from the free end of the second wire. If Y_1 and Y_2 are the Young's moduli of the wires respectively, the effective force constant of the system of two wires is

(A) $\frac{(Y_1Y_2)A}{2(Y_1L_2+Y_2L_1)}$

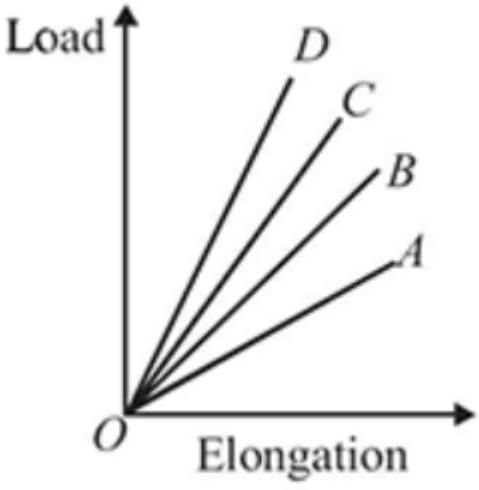
(B) $\frac{(Y_1Y_2)A}{(L_1L_2)^{1/2}}$

(C) $\frac{(Y_1Y_2)A}{Y_1L_2+Y_2L_1}$

(D) $\frac{(Y_1Y_2)^{1/2}A}{(L_2L_1)^{1/2}}$

Q.13

The load versus elongation graphs for four wires of same length and made of the same material are shown in the figure. The thinnest wire is represented by the line



(A) OA

(B) OC

(C) OD

(D) OB

Q.14

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Two wires A and B are of the same material. Their lengths are in the ratio $1:2$ and the diameter are in the ratio $2:1$. If they are pulled by the same force, then increase in length will be in the ratio

(A) $2:1$

(B) $1:4$

(C) $1:8$

(D) $8:1$

Q.15

A hollow cylindrical column of steel with inner and outer radii $20cm$ and $30cm$ supports a load of $12,000 \quad kg$. Assume the load distribution to be uniform. The value of compressional strain in terms of 10^{-6} of the column is... (Take $Y_{steel} = 2\times10^{11}$ Pa, $\pi = 3.14$, $g = 9.8$ m / s^2

(A) 3.65

(B) 3.55

(C) 3.85

(D) 3.75

Answers & Solutions

Q.1 Answer:

Both (a) \& (b)

Solution:

The deformation of a body on application of a force depends on the nature of the material and the magnitude of the applied force.

Q.2 Answer:

All of the above

Solution:

The elasticity of a material depends upon the temperature of the material. Hammering & annealing reduces elastic property of a substance.

Q.3 Answer:

$\gamma \propto 1/r^2$

Solution:

$\gamma = \frac{F}{\pi r^2} \times \frac{\ell}{\Delta\ell} \Rightarrow \gamma \propto \frac{1}{r^2}$

Q.4 Answer:

elastomers

Solution:

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The given graph does not obey Hooke's law. and there is no well defined plastic region. So the graph represents elastomers.

Q.5 Answer:

$3 : \pi$

Solution:

$\delta = \frac{\mathrm{W}\ell^3}{3\mathrm{YI}}$, **where** $\mathrm{W} = \text{load}, \ell =$ **length of beam and** I **is geometrical moment of inertia for rectangular beam,** $I = \frac{bd^3}{12}$ **where** $b =$ **breadth and** $d =$ **depth For square beam** $\mathrm{b} = \mathrm{d}$

$\therefore \mathrm{I_l} = \frac{\mathrm{b}^4}{12}$

For a beam of circular cross-section, $\mathrm{I_2} = \left(\frac{\pi \mathrm{r}^4}{4}\right) \therefore \quad \delta_1 = \frac{\mathrm{W}\ell^3 \times 12}{3\mathrm{Yb}^4} = \frac{4\,\mathrm{W}\ell^3}{\mathrm{Yb}^4}$ **(for sq. cross section) and** $\delta_2 = \frac{\mathrm{W}\ell^3}{3\mathrm{Y}(\pi \mathrm{r}^4/4)} = \frac{4\,\mathrm{W}\ell^3}{3\mathrm{Y}(\pi \mathrm{r}^4)}$

(for circular cross-section) Now $\frac{\delta_1}{\delta_2} = \frac{3\pi r^4}{b^4} = \frac{3\pi r^4}{(\pi r^2)^2} = \frac{3}{\pi}$

$\left(\because b^2 = \pi r^2\right.$ **i.e., they have same cross-sectional area)**

Q.6 Answer:

$5b - 4a$

Solution:

Using Hooke's law, $F = kx$ **we can write** $\cdots$

$$4 = k\left(a - \ell_0\right)$$

and

$$5 = k\left(b - \ell_0\right)$$

If ℓ **be the length under tension** $9\ \mathrm{N}$**, then** $\cdots$

$$9 = k\left(\ell - \ell_0\right)$$

After solving above equations, we get

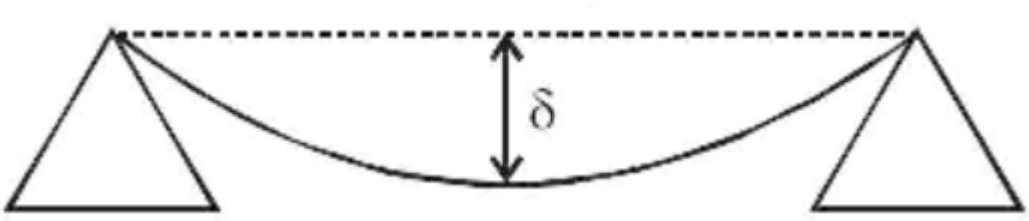
$$\ell = (5b - 4a)$$

Q.7 Answer:

$1/Y$

Solution:

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For a beam, the depression at the centre is given by,

$[\mathrm{f}, \mathrm{L}, \mathrm{b}, \mathrm{d}$ **are constants for a particular beam** $]$

i.e. $\delta \propto \frac{1}{Y}$

Q.8 Answer:

$3 : \pi$

Solution:

$\delta = \frac{\mathrm{W}\ell^3}{3\mathrm{YI}}$, **where** $\mathrm{W} = \text{load}, \ell =$ **length of beam and** I **is geometrical moment of inertia for rectangular beam,** $I = \frac{bd^3}{12}$ **where** $b =$ **breadth and** $d =$ **depth For square beam** $\mathrm{b} = \mathrm{d}$

$\therefore \mathrm{I_l} = \frac{\mathrm{b}^4}{12}$

For a beam of circular cross-section, $\mathrm{I_2} = \left(\frac{\pi \mathrm{r}^4}{4}\right) \therefore \quad \delta_1 = \frac{\mathrm{W}\ell^3 \times 12}{3\mathrm{Yb}^4} = \frac{4\,\mathrm{W}\ell^3}{\mathrm{Yb}^4}$ **(for sq. cross section) and** $\delta_2 = \frac{\mathrm{W}\ell^3}{3\mathrm{Y}(\pi \mathrm{r}^4/4)} = \frac{4\,\mathrm{W}\ell^3}{3\mathrm{Y}(\pi \mathrm{r}^4)}$

(for circular cross-section) Now $\frac{\delta_1}{\delta_2} = \frac{3\pi r^4}{b^4} = \frac{3\pi r^4}{(\pi r^2)^2} = \frac{3}{\pi}$

$\left(\because b^2 = \pi r^2\right.$ **i.e., they have same cross-sectional area)**

Q.9 Answer:

$\frac{3a}{2b^2c}$

Solution:

According to questions,

$\frac{\ell_\mathrm{s}}{\ell_\mathrm{b}} = \mathrm{a}, \frac{\mathrm{r_s}}{\mathrm{r_b}} = \mathrm{b}, \frac{\mathrm{y_s}}{\mathrm{y_b}} = \mathrm{c}, \frac{\Delta \ell \mathrm{s}}{\Delta \ell_\mathrm{b}} = ?$

As, $y = \frac{\mathrm{F}\ell}{\mathrm{A}\Delta\ell} \Rightarrow \Delta\ell = \frac{\mathrm{F}\ell}{\mathrm{Ay}}$

$\Delta\ell_\mathrm{s} = \frac{3\mathrm{mg}\ell_\mathrm{s}}{\pi \mathrm{r_s^2} \cdot \mathrm{y_s}} \left[\because \mathrm{F_s} = (\mathrm{M} + 2\mathrm{M})\mathrm{g}\right]$

$\Delta\ell_\mathrm{b} = \frac{2\mathrm{Mg}\ell_\mathrm{b}}{\pi \mathrm{r_b^2} \cdot \mathrm{y_b}} \left[\because \mathrm{F_b} = 2\mathrm{Mg}\right]$

$\therefore \frac{\Delta\ell_\mathrm{s}}{\Delta\ell_\mathrm{b}} = \frac{\frac{3\mathrm{Mg}\ell_\mathrm{s}}{\pi \mathrm{r}^2 \cdot \mathrm{y_s}}}{\frac{2\mathrm{Mg} \cdot \ell_\mathrm{b}}{\pi \mathrm{r_b^2} \cdot \mathrm{y_b}}} = \frac{3\mathrm{a}}{2\,\mathrm{b}^2\mathrm{C}}$

Q.10 Answer:

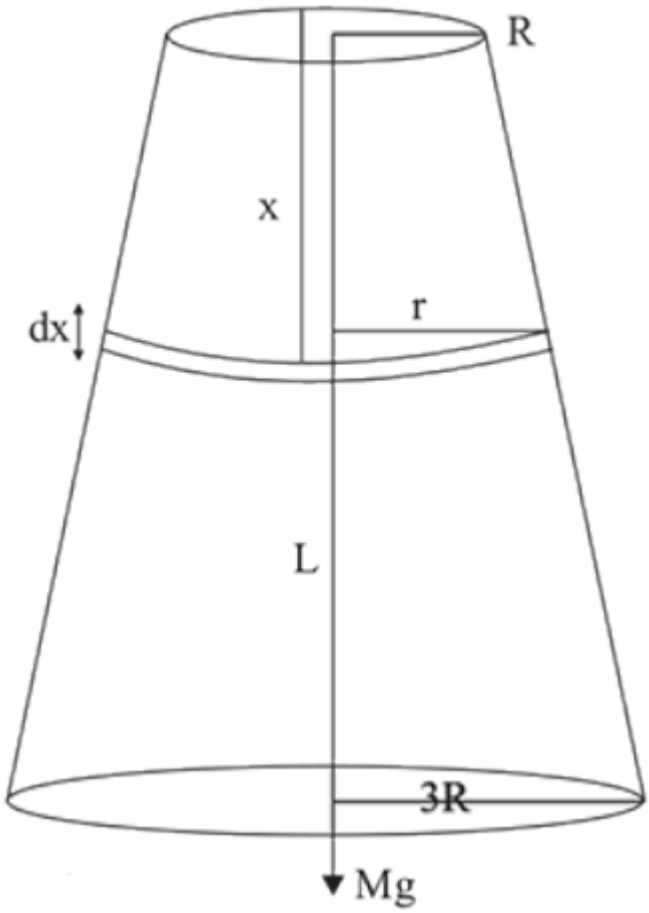
$\mathrm{L}\left(1 + \frac{1}{3}\frac{\mathrm{Mg}}{\pi \mathrm{YR}^2}\right)$

Solution:

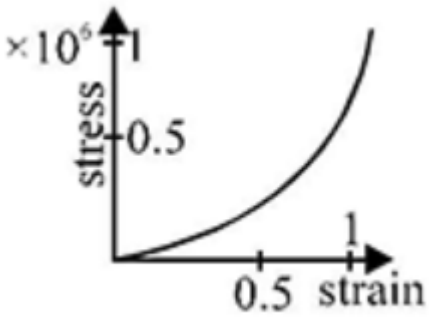
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Taking limit from 0 to L $\Delta \mathrm{L}=\frac{\mathrm{Mg}}{\pi \mathrm{y}}\left[-\frac{1}{\left[\frac{2 \mathrm{Rx}}{\mathrm{L}}+\mathrm{R}\right]_0^{\mathrm{L}}} \times \frac{\mathrm{L}}{2 \mathrm{R}}\right]=\frac{\mathrm{MgL}}{3 \pi \mathrm{R}^2 \mathrm{y}}$

The equilibrium extended length of wire $=\mathrm{L}+\Delta \mathrm{L}=\mathrm{L}+\frac{\mathrm{MgL}}{3 \pi \mathrm{R}^2 \mathrm{Y}}=\mathrm{L}\left(1+\frac{1}{3} \frac{\mathrm{Mg}}{\pi \mathrm{YR}^2}\right)$



Q.11 Answer:



Solution:

No solution available

Q.12 Answer:

$\frac{(Y_1Y_2)A}{Y_1L_2+Y_2L_1}$

Solution:

Using the usual expression for the Young's modulus, the force constant for the wire can be written as $\mathrm{k}=\frac{F}{\Delta l}=\frac{YA}{L}$ **where the symbols have their usual meanings. Now the two wires together will have an effective force constant** $\left[\frac{k_1k_2}{k_1+k_2}\right]$ **. Substituting the corresponding lengths and the Young's moduli we get the answer.**

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Q.13 Answer:

OA

Solution:

Using $\mathrm{Y} = \frac{\mathrm{F/A}}{\Delta \mathrm{L/L}}$

Cross-section area of the wire $\mathrm{A} = \pi \mathrm{R}^2$ **where** R **is the radius of the wire**

$\therefore \quad \mathrm{Y} = \frac{\mathrm{F}}{\pi \mathrm{R}^2} \times \frac{\mathrm{L}}{\Delta \mathrm{L}} \quad \Longrightarrow \quad \mathrm{F} = \frac{\pi \mathrm{R}^2 \mathrm{Y}}{\mathrm{L}} \Delta \mathrm{L}$

Thus Slope of load vs elongation graph, $\mathrm{s} = \frac{\pi \mathrm{R}^2 \mathrm{Y}}{\mathrm{L}} \Longrightarrow \quad \mathrm{s} \propto \mathrm{R}^2$

Now from the graph shown, OA has the least slope and hence the least radius. Thus OA represents the thinnest wire.

Q.14 Answer:

$1 : 8$

Solution:

We know that Young's modulus $\mathrm{Y} = \frac{\mathrm{F}}{\pi \mathrm{r}^2} \times \frac{\mathrm{L}}{\ell}$

Since Y, F **are same for both the wires, we have,**

$\frac{1}{\mathrm{r}_1^2} \frac{\mathrm{L}_1}{\ell_1} = \frac{1}{\mathrm{r}_2^2} \frac{\mathrm{L}_2}{\ell_2}$

or, $\frac{\ell_1}{\ell_2} = \frac{\mathrm{r}_2^2 \times \mathrm{L}_1}{\mathrm{r}_1^2 \times \mathrm{L}_2} = \frac{(\mathrm{D}_2/2)^2 \times \mathrm{L}_1}{(\mathrm{D}_1/2)^2 \times \mathrm{L}_2}$

or, $\frac{\ell_1}{\ell_2} = \frac{\mathrm{D}_2^2 \times \mathrm{L}_1}{\mathrm{D}_1^2 \times \mathrm{L}_2} = \frac{\mathrm{D}_2^2}{(2\mathrm{D}_2)^2} \times \frac{\mathrm{L}_2}{2\,\mathrm{L}_2} = \frac{1}{8}$

So, $\ell_1 : \ell_2 = 1 : 8$

Q.15 Answer:

3.75

Solution:

Area of cross-section of the cylindrical column,

$\mathrm{a} = \pi \mathrm{r}_2^2 - \mathrm{r}_1^2 = \pi 0.3^2 - 0.2^2 = 0.05 \pi \mathrm{m}^2$

Mass supported on the column, $\mathrm{M} = 12,000 \mathrm{kg}$ **Therefore, compressional force on the column,**

$\mathrm{F} = \mathrm{Mg} = 12,000 \times 9.8 = 117600 \mathrm{N}$

Compressional strain,

$\frac{l}{\mathrm{L}} = \frac{\mathrm{F}}{\mathrm{aY}} = \frac{117600}{0.05\pi \times 2 \times 10^{11}} = \frac{117600}{3.14 \times 10^{10}}$

$= 3.7452 \times 10^{-6}$

$= 3.75 \times 10^{-6}$ **....(Rounding off to 2 decimal places.)**

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